

Continuous Space-Time Video Resampling with Invertible Motion Steganography

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Abstract

Space-time video resampling aims to conduct both spatial-temporal downsampling and upsampling processes to achieve high-quality video reconstruction. Although there has been much progress, some major challenges still exist, such as how to preserve motion information during temporal resampling while avoiding blurring artifacts, and how to achieve flexible temporal and spatial resampling factors. In this paper, we introduce an Invertible Motion Steganography Module (IMSM), designed to embed motion information from high-frame-rate videos into downsampled frames with lower frame rates in a visually imperceptible manner. Its reversible nature allows the motion information to be recovered, facilitating the reconstruction of high-frame-rate videos. Furthermore, we propose a 3D implicit feature modulation technique that enables continuous spatiotemporal resampling. With tailored training strategies, our method supports flexible frame rate conversions, including non-integer changes like 30 FPS to 24 FPS and vice versa. Extensive experiments show that our method significantly outperforms existing solutions across multiple datasets in various video resampling tasks with high flexibility. Codes will be made available at the URL <https://github.com/hahazh/CSTVR>.

1. Introduction

Spatial and temporal downsampling are widely applied in video processing to ensure compatibility across various devices and playback settings, facilitate storage [45, 54], and enable efficient transmission and preview [38]. Additionally, certain video coding standards [59] and high-level computer vision tasks [6, 45, 47] utilize space-time downsampling to reduce complexity. However, reconstructing high-fidelity video from downsampled versions is challenging due to the inherently ill-posed nature of the problem.

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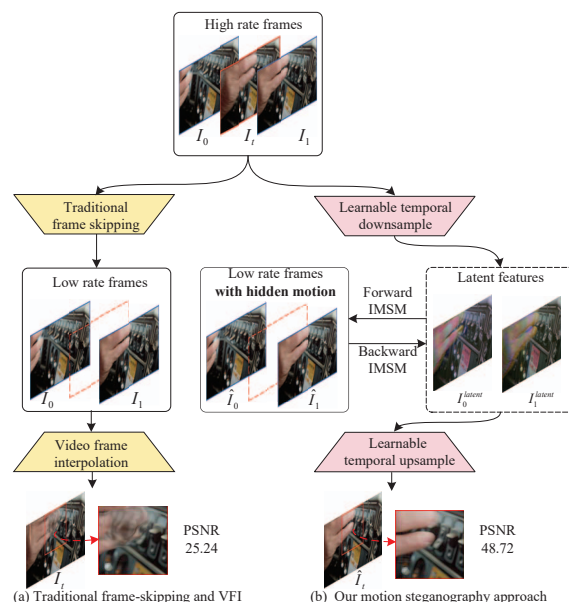


Figure 1. (a) Traditional "frame-skipping and VFI": Motion information is discarded, causing blur in reconstructed frames. (b) Our motion steganography approach: The IMSM reversibly embeds motion information into the low-frame-rate results in a imperceptible manner, aiding in frame reconstruction.

Given these practical considerations, video resampling aims to jointly optimize the downsampling and upsampling processes to achieve better resampling results. Numerous techniques for image [14, 38, 43, 52, 54] and video resampling [10, 15, 45] have been developed, aiming to enhance the performance by optimizing both the upsampling and downsampling procedures. Despite years of progress, two key challenges in video resampling remain underexplored: (1) **effective modeling of temporal resampling** and (2) **continuous space-time resampling**. Regarding the first issue, most existing methods primarily focus on spatial resampling, with less emphasis on temporal resampling. As spatial downsampling does not require explicit strict requirements; various methods can be chosen as long as they

meet criteria such as preserving important features and reducing aliasing effects. Techniques like bilinear, bicubic, and learning-based methods are viable options. In contrast, temporal downsampling poses stricter constraints. Uniform frame subsampling, the most common approach, maintains motion consistency but at the cost of discarding vital motion information. Attempts to blend frames or use non-uniform sampling often introduce blurring artifacts and compromise the visual quality, especially in dynamic scenes [39]. Moreover, recovering high-frame-rate video from temporally downsampled frames is further complicated as the motion information discarded during the subsampling process is nearly impossible to recover perfectly. Despite advancements in video frame interpolation (VFI) techniques [8, 16, 22, 58], reconstructing complex motion remains difficult. While STAA [50] makes progress in mitigating some of these issues, it still struggles to effectively preserve the motion information lost in high-frame-rate videos and lacks a comprehensive evaluation of the quality of downsampled images. To address these limitations, we propose an Invertible Motion Steganography Module (IMSM) that embeds motion information within low-frame-rate videos. Unlike conventional methods that either compromise the subjective quality of the downsampled video [18, 37, 60] or fail to preserve motion information altogether, our approach strikes a balance. The IMSM generates results that visually match uniform frame subsampling, ensuring that the downsampled frames are free from motion blur and retain consistent motion appearance. Meanwhile, this technique embeds motion information in a reversible manner, supporting the reconstruction of high-frame-rate videos with enhanced fidelity. Figure 1 contrasts our approach with traditional methods, highlighting the benefits of motion preservation through steganography. We also conduct extensive experiments, including objective evaluations and subjective tests, to assess the quality of the downsampling results.

Regarding the second issue, existing video resampling models are limited to fixed temporal and spatial resampling. While some space-time video super-resolution methods [4, 5, 64, 65] have achieved controllable spatiotemporal upsampling, they do not consider the joint optimization of downsampling and upsampling processes. Moreover, these methods only perform integer-based temporal upsampling (e.g., 30 FPS to 60 FPS) and cannot handle fractional frame rate transformations (e.g., 24 FPS to 30 FPS). To address this issue, we propose a straightforward yet effective spatiotemporal implicit modulation module for continuous upsampling and downsampling. This method can be seen as an extension of implicit function-based image super-resolution methods [3, 24]. Equipped with invertible motion steganography and spatiotemporal implicit modulation, our approach achieves high flexibility and improved performance. Our contributions are summarized as

follows:

- To address the issue of irrecoverable motion information lost during temporal downsampling, we introduce an invertible motion steganography module that imperceptibly embeds motion information from high-frame-rate videos into low-frame-rate frames. Its reversible nature allows for the restoration of this motion information, thereby aiding in high-quality spatiotemporal reconstruction.
- To the best of our knowledge, this is the first work to achieve continuous spatiotemporal resampling. We propose a straightforward yet effective 3D implicit modulation module, along with a corresponding training strategy, to enable continuous spatiotemporal modulation, including non-integer frame rate conversions.
- Extensive experiments show that the proposed method excels in various video restoration tasks, including temporal resampling, spatial resampling, and spatiotemporal resampling, demonstrating its flexibility and high performance.

2. Related Work

2.1. Image / Video Resampling

Image resampling aims to jointly optimize both downsampling and upsampling processes to achieve high-quality reconstruction. Some earlier methods [21, 26, 52] achieve image resampling by simultaneously optimizing the parameters of both upsampling and downsampling models. Current research efforts have mainly concentrated on refining the resampling procedure [25, 36, 53], accounting for compression artifacts [38, 54], accommodating asymmetric upsampling and downsampling [35]. We note that FGRN [25] uses a seemingly similar approach by incorporating an additional invertible neural network for single-image rescaling; however its motivation, and its functionality differ fundamentally from our work, as discussed in the supplementary material. Video resampling task has received increased attention in recent years. Tian *et al.* [45] introduce a self-conditioned probabilistic framework for video resampling that jointly learns the upsampling and downsampling procedures. Huang *et al.* [15] present two joint optimization approaches based on invertible neural networks with coupling layers and discuss different upsampling and downsampling strategies. To our knowledge, STAA [50] is the only work that addresses both spatial and temporal resampling using an end-to-end trained autoencoder framework. However, this approach, which relies solely on unsupervised methods for spatiotemporal low-pass filtering, can result in blurriness in downsampled frames, particularly in the presence of significant motion. In contrast, our method can reversibly hide motion information without compromising the subjective quality of the downsampled frames.

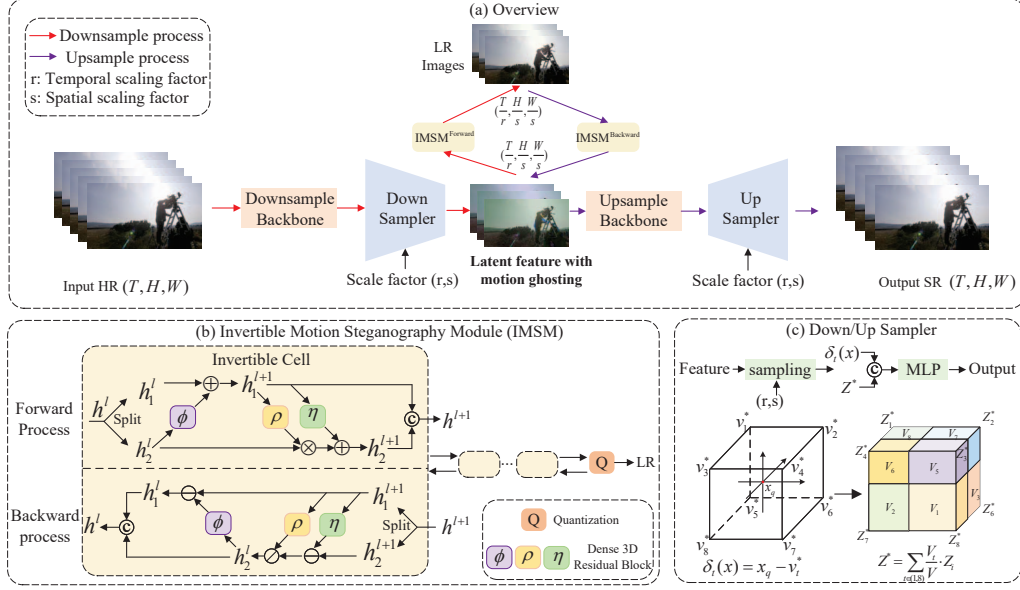


Figure 2. (a) The architecture and data flow of the proposed approach. Downsampling process: The input HR frames are first downsampled to obtain latent features with motion ghosting, which are then passed through the IMSM to generate LR images (indicated by red arrows). Upsampling process: The LR images are first converted back to latent features with inverse transformation. Then, multiple frames are mutually enhanced and upsampled to restore the SR output (indicated by purple arrows). (b) The Invertible Motion Steganography Module (IMSM) is composed of invertible flow cells, which can embed motion information into LR images and reverse the process when needed. (c) The Down/Up sampler modulates the input features to the desired shape using 3D local implicit modulation.

2.2. Image Steganography

Image steganography aims to hide data within an image container. Historically, the least significant bit [19] has been widely employed in steganographic techniques. Other spatial domain data hiding methods include pixel-value differencing based techniques [48, 49] and palette shifting-based methods [17, 34]. In recent years, learning-based approaches have rapidly emerged. Prominent methods involve the use of Generative Adversarial Networks (GANs) for synthesizing container images [40, 61] and some normalizing flow-based techniques [33, 55]. There are also works in the field of image restoration that have adopted steganography techniques. For example, ISRN [13] employs steganography to address the problem of single-image rescaling, while IICNet [7] designs a flow-based invertible neural network to achieve efficient video embedding. However, the rationale and specific implementation strategies of these methods differ significantly from our approach. To our knowledge, this is the first work to apply steganography to continuous space-time video restoration.

3. Methodology

3.1. Problem Formulation

Given a sequence of frames, $F = \{I_i\}_1^T$, where each frame $I_i \in \mathbb{R}^{s \cdot H \times s \cdot W \times 3}$ with a spatial resampling factor s and a

temporal resampling factor r , our goal is to design a jointly optimized downsampler $\mathcal{D}$ and upsampler $\mathcal{U}$. The downsampler outputs a sequence of downsampled frames $F_d = \{\mathcal{D}(I_i)\}_1^T$, where $\mathcal{D}(I_i) \in \mathbb{R}^{H \times W \times 3}$. These downsampled frames should maintain similar subjective visual quality as traditional downsampling results. When needed, the upsampler can re-construct the original high-resolution and high-frame-rate sequence, denoted as $\hat{F} = \{\mathcal{U}(\mathcal{D}(I_i))\}_1^T$. The reconstructed video $\hat{F}$ should strive to be as close as possible to the original video F .

3.2. System Overview

The overview of our method is shown in Figure 2(a). Our model consists of two main components: an end-to-end optimized resampler, which includes a sequential downsampler and upsampler, and an Invertible Motion Steganography Module (IMSM). These two components also correspond to the two training stages. In the first stage, we optimize the resampler end-to-end, supervising only the output SR while applying no supervision to the downsampled results. During this process, the downsampler encodes the input into latent features matching the dimensions of the downsampled frames, and the upsampler reconstructs SR frames by upsampling these latent features. This process encourages the latent features to retain as much motion information as possible, often resulting in noticeable **motion ghosting artifacts**. In the second stage, we introduce the

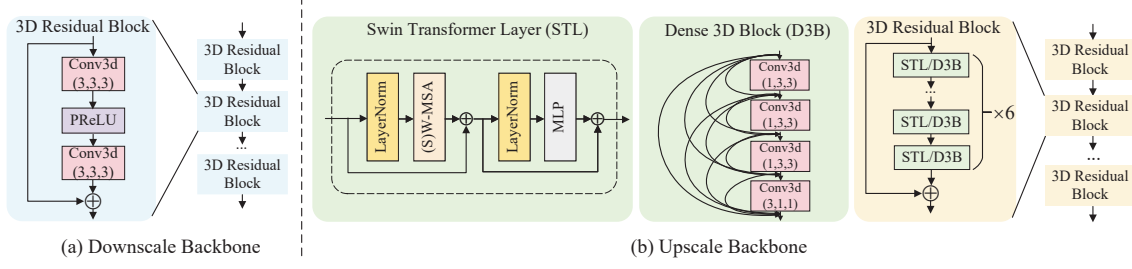


Figure 3. (a) The structure of the Downscale Backbone which includes multiple 3D Residual blocks. (b) The structure of the Upscale Backbone. We use two modules as the basic building blocks for the upsampling part: the Swin Transformer Layer and the Dense 3D Block. These two modules are then stacked together to form the 3D Residual blocks used for upsampling.

IMSM. This module takes the latent features as input and converts them into visually pleasing RGB images that embed motion information. When needed, the IMSM can invert these RGB images back into latent features for upsampling. Specific details of the model design are provided in the relevant sections.

3.3. Resample Backbone

3.3.1 Downsample Backbone

Inspired by recent advances in video super-resolution [1, 2], as illustrated in Figure 3(a), we use multiple residual blocks as the backbone of the downsampler. Specifically, each residual block is constructed using a Conv3d-PReLU-Conv3d structure, with Conv3d having a kernel size of $3 \times 3 \times 3$.

3.3.2 Upsample Backbone

We use two types of 3D modules as feature extractors for the upsampling component, tailored to different scenarios. When both spatial and temporal resolutions are altered, we employ the 3D Swin Transformer Layer as the basic block (denoted as STL in Figure 3(b)). When only temporal resampling is involved and the spatial resolution remains unchanged, we choose a pure convolutional block as the basic block (denoted as D3B in Figure 3(b)) to reduce GPU memory consumption for high-resolution input images.

3D Swin Transformer Layer. Inspired by recent advancements in image restoration [9, 27, 30] and high-level video understanding methods [29, 57], we propose an extension of the 2D Swin block that operates on spatiotemporal patches to address space-time restoration tasks. Given a video sequence with dimensions $T \times H \times W$, we define the spatiotemporal window size as $n \times m \times m$, dividing the video into $\lceil \frac{T}{n} \rceil \times \lceil \frac{H}{m} \rceil \times \lceil \frac{W}{m} \rceil$ non-overlapping windows. Standard self-attention is then computed within each window. For a feature patch P of size $n \cdot m^2$, the query, key, and value matrices Q , K , and V are obtained by multiplying P with their respective linear projection matrices W_Q , W_K , and

W_V , each in $\mathbb{R}^{n \cdot m^2 \cdot d}$. The attention output is calculated as $\text{atten}(Q, K, V) = \text{SoftMax}(\frac{QK^T}{\sqrt{d}} + B)V$, where B represents a learnable positional encoding parameter. In practice, multi-head self-attention is applied in parallel across each window.

Dense 3D Block. We extend the Residual in Residual approach [62] from spatial dimensions to a 3D spatiotemporal dense block. Illustrated in Figure 3(b), to streamline computational complexity, we adopt depth-wise separable convolution. The initial convolution kernels are configured as $1 \times 3 \times 3$, while the final convolution kernel is set as $3 \times 3 \times 1$, facilitating the extraction of spatiotemporal information.

3.4. Space-Time Modulation Sampler

To enable continuous modulation of spatial resolution, certain image super-resolution techniques [3, 24] utilize 2D implicit functions for image representation. In this study, we extend these 2D implicit image functions to incorporate a temporal dimension, thereby creating a 3D implicit function. The downsampler and upsampler in our method have identical structures but do not share weights. The downsampler converts spatiotemporal features into a specified downsampled latent feature size, while the upsampler modulates the downsampled features back to their original dimensions. Given the shared structure, we use the downsampler as an illustrative example. As shown in Figure 2(c), in our 3D implicit representation, the intermediate spatiotemporal latent feature is encapsulated in a 3D feature map $z \in \mathbb{R}^{T \times H \times W}$ (the channel axis is omitted for simplicity). A decoding function g_θ , parameterized by θ , is applied to modulate z to the corresponding downsampled scale, as expressed below:

$$s = g_\theta(z, x), \quad (1)$$

where $x \in \mathcal{X}$ is a 3D coordinate in the continuous spatiotemporal domain, and $s \in \mathcal{S}$ is the predicted value. Under this definition, each input vector z can be viewed as representing an implicit spatiotemporal function $g_\theta(z, \cdot) : \mathcal{X} \rightarrow \mathcal{S}$. To achieve a continuous spatiotemporal representation, we model each modulated feature point as a

weighted sum of the values from its eight nearest neighboring spatiotemporal positions. For the continuous latent feature $L^i(x_q)$, the modulated value at coordinate x_q can be formulated as:

$$L^i(x_q) = \sum_{t \in (1,8)} \frac{V_t}{V} \cdot g_\theta(z_t^*, \delta_t(x)), \quad \delta_t(x) = x_q - v_t^*, \quad (2)$$

where z_t^* refers to the nearest eight corner points of a space-time cube, and v_t^* represents the 3D coordinates of z_t^* . V_t signifies the volume of the cuboid between the query point x_q and the diagonal corner (e.g., the front top left corner corresponding to the back bottom right cubic). Conceptually, this is similar to bilinear sampling, where the closer a point is to a corner, the greater its corresponding weight. The weights are normalized by $V = \sum_t V_t$. Subsequently, the relative positional coordinates and their respective weighted sum values are combined and input into a multi-layer perceptron to derive the final output value. The upsampling and downsampling components have identical structures, differing only in that the upsampling unit is designed to convert features into RGB values, while the downsampling unit's output latent features.

3.5. Invertible Motion Steganography Module

Since motion information cannot be restored once lost, preserving it is essential for restoring high-frame-rate videos. Traditional methods either discard motion information, resulting in poor reconstruction, or introduce temporal blur [18, 37, 50, 60], which affects the visual quality of low-frame-rate videos. To address this issue, we draw inspiration from image steganography and propose a strategy to embed motion information in a visually imperceptible manner within low-frame-rate downsampled frames. This approach allows us to retain key motion information for upsampling without sacrificing the quality of the downsampled frames. As mentioned earlier in Section 3.2, during the first stage of training, we do not impose any constraints on the intermediate downsampled features. Consequently, the low-frame-rate latent features often exhibit **motion ghosting artifacts** (as shown in Figure 2(a)). To address this, we use an invertible structure to hide and restore motion information using a single network. Our invertible motion steganography module consists of several reversible blocks, as shown in Figure 2(b). We use enhanced affine transformation [11] as fundamental building blocks. Each l^{th} invertible block divides the input h^l into two parts, h_1^l and h_2^l , along the channel axis. The corresponding outputs are h_1^{l+1} and h_2^{l+1} . The forward operation is formulated as:

$$\begin{aligned} h_1^{l+1} &= h_1^l + \phi(h_2^l), \\ h_2^{l+1} &= h_2^l \otimes \exp(\rho(h_1^{l+1})) + \eta(h_1^{l+1}), \end{aligned} \quad (3)$$

where $\phi(\cdot)$ and $\eta(\cdot)$ are arbitrary functions, $\exp(\cdot)$ and $\otimes$ are exponential function and Hadamard product respectively. We utilize dense 3D blocks as functions to extract spatiotemporal features from multiple frames. For the backward operation, given $[h_1^{l+1}, h_2^{l+1}]$, we can inversely calculate $[h_1^l, h_2^l]$ as:

$$\begin{aligned} h_2^l &= (h_2^{l+1} - \eta(h_1^{l+1})) \otimes \exp(-\rho(h_1^{l+1})), \\ h_1^l &= h_1^{l+1} - \phi(h_2^l). \end{aligned} \quad (4)$$

For the IMSM module, the forward process converts latent features with ghosting artifacts into visually pleasing LR images with hidden motion information. The backward process, in turn, converts the visually pleasing LR images back into latent features for upsampling. The training process consists of two stages. In the first stage, we train the resampling model using the Charbonnier loss [23] in an end-to-end manner, supervising only the output of the upsampler. The second training stage is to develop an IMSM capable of transforming motion-blurred latent features into visually pleasing RGB images. Since the stored RGB images require quantization, complete reversibility cannot be achieved, and the quantization operation is non-differentiable. Here, we adopt Differentiable Quantization Layer in [50] to approximate gradient for backpropagation. To ensure robustness to quantization losses in both conversion directions, we concurrently supervised the forward and backward operations of the reversible neural network. The formulation of the invertible loss is as follows:

$$loss_{inv} = \underbrace{L_{char}(\Phi(I_{la}), I_{bic})}_{\text{forward process}} + \underbrace{L_{char}(\Phi^{-1}(\Phi(I_{la})), I_{la})}_{\text{backward process}} \quad (5)$$

Here, Φ and Φ^{-1} represent the forward and backward processes of the invertible motion steganography module. I_{la} and I_{bic} denote the latent feature and the corresponding low-resolution, low-frame-rate images, respectively. Bicubic and nearest-neighbor downsampling methods are used for spatial and temporal downsampling guidance.

4. Experiments

4.1. Experimental Setup

We use the Vimeo90k dataset [56], a standard dataset for video restoration tasks, to train models at fixed scales. Specifically, we train two models corresponding to fixed spatiotemporal sampling scales: (1) $2 \times$ temporal resampling and (2) $2 \times$ temporal and $4 \times$ spatial resampling. Following established practices [50], we evaluate the algorithm's performance using the Vimeo90k test set and Vid4 [28]. To achieve continuous spatiotemporal resampling, we utilize high-frame-rate video frames from Adobe240FPS [42] as the training set and evaluate continuous spatiotemporal super-resolution on the SPMCS dataset [44]. We use

Table 1. Comparisons with State-of-the-Art video frame interpolation methods, space-time video super-resolution methods, and space-time resampling methods on benchmark datasets, Metrics: PSNR(dB)/SSIM. The best and the second-best results are in red and blue, respectively. The results of STAA are directly quoted from the original paper, as there is no available open-source code.

Upscale rate time/space	Downsampler time/space	Reconstruction method	Params/M	Vimeo-90k		Vid4	
				RGB-Space	Y-Space	RGB-Space	Y-Space
$2\times / 1\times$	Nearest/-	XVFI [41]	5.7	34.16/0.9456	36.16/0.9603	28.50/0.9360	30.26/0.9477
		FLAVR [20]	42.1	36.27/0.9599	38.38/0.9716	29.39/0.9361	31.17/0.9467
		EDSC [8]	8.9	34.48/0.9489	36.35/0.9616	29.20/0.9372	31.05/0.9491
		IFRNet [22]	19.7	35.22/0.9544	37.12/0.9668	29.55/0.9406	31.39/0.9519
		RIFE [16]	20.9	35.25/0.9535	37.28/0.9662	28.41/0.9335	30.15/0.9454
		EMA-VFI [58]	65.7	36.08/0.9585	38.02/0.9697	29.80/0.9434	31.62/0.9538
		VFIFormer [31]	24.1	35.93/0.9581	37.88/0.9689	28.94/0.9400	30.70/0.9506
	STAA[50]	STAA [50]	15.9	45.01/0.9912	-	39.78/0.9926	-
	Ours	Ours	11.7	48.03/0.9949	50.52/0.9976	43.75/0.9955	46.77/0.9974
	$2\times / 4\times$	Nearest/Bicubic	EDSC [8]+BasicVSR++ [2]	8.9+7.3	32.79/0.9092	34.51/0.9252	24.48/0.7572
IFRNet [22]+BasicVSR++ [2]			19.7+7.3	32.88/0.9098	34.70/0.9268	24.52/0.7568	26.36/0.7917
RIFE [16]+BasicVSR++ [2]			20.9+7.3	32.81/0.9091	34.56/0.9253	24.51/0.7582	26.37/0.7939
ZSM [51]			11.1	33.48/0.9178	35.21/0.9323	24.82/0.7763	26.31/0.7976
STDAN [46]			8.3	33.59/0.9192	35.50/0.9351	24.91/0.7832	26.28/0.8041
MoTIF [4]			12.6	32.31/0.9098	34.03/0.9265	24.26/0.7527	25.79/0.7752
RSTT [12]			7.7	33.66/0.9192	35.39/0.9336	24.94/0.7805	26.43/0.7994
OFR-BRN [63]			11.77	33.71/0.9206	35.53/0.9358	25.21/0.7957	26.72/0.8141
MsMr [64]			13.19	33.76/0.9214	35.57/0.9365	25.41/0.8044	26.87/0.8213
STAA [50]			STAA [50]	16.0	34.53/0.9426	-	27.31/0.9173
Ours	Ours	17.4	37.78/0.9624	40.57/0.9770	30.71/0.9336	32.76/0.9488	

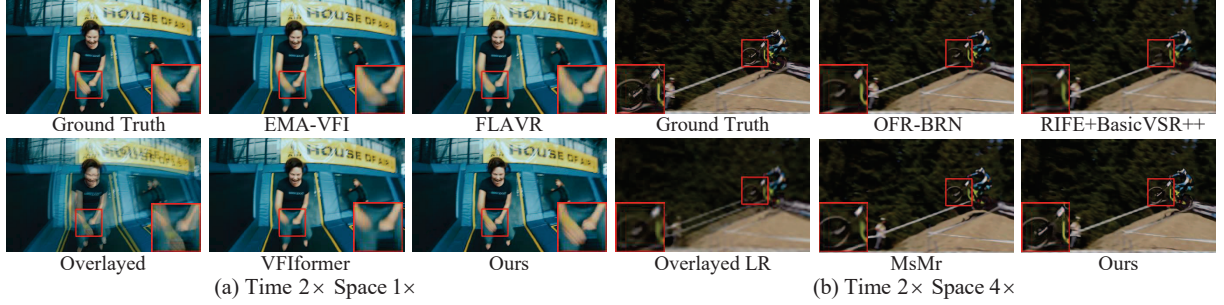


Figure 4. Visual qualitative comparisons on reconstruction of different methods on Vimeo dataset.

the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ for training. Details of the loss functions have been discussed in Section 3.5, and a full description of the experimental setup is provided in the **supplementary material**.

4.2. Comparison with State-of-the-Art Methods

Fixed Space-time Resampling. We follow the settings in [50] and evaluate different combinations of time and space sampling rates. Our comparison includes Time 2× Space 1× as well as Time 2× Space 4×. The quantitative results are presented in Table 1. For the Time 2× Space 1× case, we compare our approach with several video frame interpolation algorithms [8, 16, 20, 22, 31, 41, 58] and joint optimized resampling method STAA [50]. Given the varied testing conditions and color spaces used across different methods, we provide the results in both RGB and YCbCr spaces. For the Time 2×, Space 4× case, the comparison algorithms include sequentially applying video frame interpolation [8, 16, 22] and video super-resolution tech-

niques [2] and one-stage space-time video super-resolution methods [4, 12, 46, 51, 63, 64]. As shown in Table 1, our approach significantly outperforms all other methods across all spatiotemporal combinations. Compared to the second-best frame interpolation method, our technique achieves an improvement of approximately 13 dB on the Vimeo90k dataset and around 16 dB on the Vid4 dataset. Similarly, when compared with STAA, which also uses joint downsampling and upsampling optimization, our method demonstrates substantial advantages, surpassing STAA by 3 dB on Vimeo90k and by 4 dB on Vid4. In comparisons with space-time video super-resolution methods, our approach yields similar conclusions, showing improvements of over 3 dB compared to the second-best solution on both datasets. These results validate the effectiveness of the proposed approach. Subjective visual comparisons are provided in Figure 4, showing that video interpolation methods tend to introduce noticeable blurring when significant nonlinear motion exists between adjacent frames. In contrast, our proposed approach can reconstruct

Table 2. Quantitative Comparison of different methods for non-integer sampling rates on SPMCS [44], Metrics: PSNR(dB)/SSIM. The best and the second-best results are in red and blue, respectively.

Methods	S=2.0		S=2.4		S=2.8	
	RGB-Space	Y-Space	RGB-Space	Y-Space	RGB-Space	Y-Space
VideoINR [5]	29.16/0.8884	30.67/0.9024	28.41/0.8886	29.86/0.9023	28.96/0.8862	30.44/0.9001
MoTIF [4]	34.06/0.9572	35.64/0.9663	31.19/0.9064	32.73/0.9230	30.13/0.8837	31.78/0.9175
MsMr [65]	34.11/0.9573	35.94/0.9672	32.62/0.9403	34.33/0.9504	31.37/0.9192	33.02/0.9308
Ours	35.74/0.9583	38.35/0.9703	35.03/0.9536	37.59/0.9665	34.17/0.9475	36.64/0.9615

Methods	S=3.2		S=3.6		S=4.0	
	RGB-Space	Y-Space	RGB-Space	Y-Space	RGB-Space	Y-Space
VideoINR [5]	28.04/0.8631	29.52/0.8772	27.19/0.8450	28.63/0.8622	27.15/0.8247	28.61/0.8435
MoTIF [4]	29.96/0.8827	31.48/0.9123	27.86/0.8401	29.31/0.8764	27.34/0.8273	28.94/0.8497
MsMr [65]	30.37/0.8959	31.98/0.9093	28.67/0.8671	30.20/0.8831	28.27/0.8424	29.80/0.8600
Ours	33.36/0.9404	35.74/0.9554	32.56/0.9317	34.87/0.9479	31.87/0.9223	34.11/0.9398

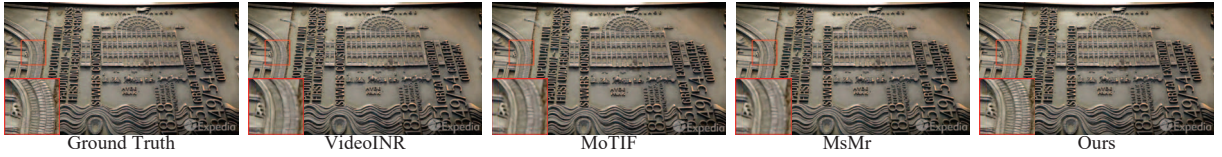


Figure 5. Visual qualitative comparisons of non-integer downsampling fatcor (3.6) for each method on the SPMCS dataset.

Table 3. Ablation study of w/wo Invertible Motion Steganography Module(IMSM) on benchmark dataset, Metrics: PSNR-Y(dB)/SSIM-Y.

Upscale rate	Vimeo90k		Vid4	
	w/o IMSM	w/ IMSM	w/o IMSM	w/ IMSM
T×2 S×4	38.86/0.9661	40.57/0.9770	30.61/0.9161	32.76/0.9488
T×2 S×1	43.76/0.9878	50.52/0.9976	44.45/0.9925	46.77/0.9974

results that are visually nearly lossless and close to the ground truth.

Continuous Space-time Resampling. We then proceed to compare continuous spatiotemporal video restoration. To the best of our knowledge, there are no learning-based methods capable of achieving non-integer temporal modulation. Therefore, we select several methods capable of continuous spatial modulation for comparison. Specifically, we keep the temporal sampling factor at 2 and varied the spatial sampling factor. The quantitative results are presented in Table 2, where our method consistently outperforms the comparison algorithms at every scale. Qualitative comparisons are shown in Figure 5, highlighting our approach’s ability to reconstruct results with sharper edges and enhanced visual appeal. Additional discussions about non-integer frame rate conversion and visual comparisons are provided in the **supplementary file**.

4.3. Ablation Study

Invertible Motion Steganography. Compared to using IMSM for embedding motion information, a more intuitive approach is to directly supervise the output of the down-

sampler, which is commonly adopted by most rescaling algorithms. To verify the effectiveness of IMSM, we remove the IMSM and simultaneously supervised the output of both the downsampler and the upsampler. We apply interval sampling for temporal downsampling guidance and bicubic sampling for spatial downsampling guidance. The quantitative results of the ablation experiments are presented in Table 3, showing that the upsampled results of models incorporating the IMSM module outperform those of models with directly supervised downsampled frames for every spatiotemporal sampling combination. Furthermore, we observe a close correlation between the improvement in performance and the characteristics of the dataset itself. Specifically, the introduction of IMSM leads to a greater performance increase in Vimeo90k compared to Vid4. This difference can be attributed to the more substantial and dynamic motion present in Vimeo90k compared to the gentler motion in Vid4.

Trade-off Between Performance and Complexity. To investigate the relationship between model complexity and performance, we adjusted the model complexity by varying the number of basic modules in the upsampling backbone. The quantitative results, presented in Table 4, demonstrate that increasing the model’s depth improves performance. It also indicates that both transformer-based and simple CNN architectures can achieve good performance. This suggests that our resampling framework demonstrates considerable robustness, being effective for both transformer and CNN architectures. Considering the quadratic complexity of the Swin Transformer block, high input resolutions may lead to GPU memory overflow. Therefore, we employed different

Table 4. Ablation experiments on model performance under different complexities, Metrics: PSNR-Y(dB)/SSIM-Y.

Downsampling Rate	block num	Vimeo90k		Vid4	
		PSNR-Y	SSIM-Y	PSNR-Y	SSIM-Y
$T \times 2 S \times 4$	12	39.52/0.9715		31.55/0.9338	
	18	39.93/0.9733		31.80/0.9378	
	24	40.57/0.9770		32.76/0.9488	
$T \times 2 S \times 1$	12	49.42/0.9970		42.63/0.9933	
	18	49.78/0.9971		46.08/0.9971	
	24	50.58/0.9977		46.57/0.9974	

Table 5. Quantitative comparison of downsampled image and its corresponding counterpart with traditional downsampling method (metrics: PSNR-Y/SSIM-Y). We apply interval sampling for temporal downsample and bicubic sampling for spatial downsample.

Downsampling Rate	Vimeo-Fast	Vimeo-Medium	Vimeo-Slow	Vimeo-Total
$T \times 2 S \times 4$	44.52/0.9955	44.68/0.9953	44.53/0.9955	44.67/0.9952
$T \times 2 S \times 1$	48.01/0.9941	47.88/0.9950	48.46/0.9962	48.03/0.9951

Table 6. No reference quantitative comparison of downsampled image and its corresponding counterpart with traditional downsampling method. (metrics: NIQE ↓).

Downsampling Rate	Vimeo-Fast		Vimeo-Medium		Vimeo-Slow		Vimeo-Total	
	Bicubic	Ours	Bicubic	Ours	Bicubic	Ours	Bicubic	Ours
$T \times 2 S \times 4$	11.91	11.65	14.27	12.17	6.47	6.46	12.29	10.91
$T \times 2 S \times 1$	6.80	6.45	8.10	8.78	4.88	5.02	7.23	7.64

backbones for varying sampling rates. For cases involving only temporal sampling ($T \times 2$, $S \times 1$), we utilized only the Dense 3D CNN block as the basic module.

4.4. Evaluation of Downsampled Images.

The main difference between video downsampling and video compression is that video downsampling aims to produce results that visually resemble those of traditional downsampling methods. To assess the quality of downsampling results, we use interval frame sampling as a reference for learned temporal downsampling and bicubic downsampling as a reference for learned spatial downsampling. The experimental results are shown in Table 5. For $T \times 2$, $S \times 4$, the PSNR is approximately 45 dB, while for $T \times 2$, $S \times 1$, it exceeds 48 dB, indicating that the visual quality of learning-based spatiotemporal downsampling closely aligns with its traditional counterpart. We also include a no-reference quality metric, NIQE [32], in Table 6, which shows results comparable to bicubic and temporal interval sampling, confirming that motion information is retained without perceptual quality loss. Further discussions and subjective experiments are provided in the **supplementary file**.

4.5. Analysis of Motion Steganography

Since visual comparisons cannot reveal perceptible differences between the downsampled frames generated by our method and those produced by traditional methods, we em-

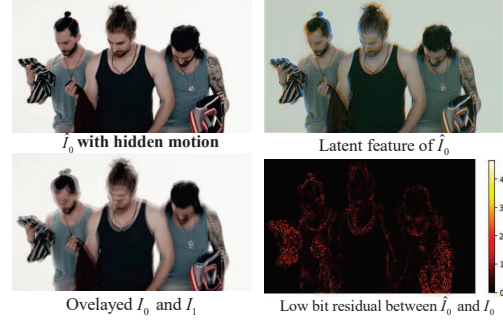


Figure 6. Visualization of Low-Bit steganographic motion.

ploy least significant bit steganography [19] to analyze the hidden motion information. Specifically, we examine the cases of $T \times 2$ and $S \times 1$. Here, for two consecutive frames I_0 and I_1 , I_0 is retained as the result of downsampling, while I_1 is removed. IMSM embeds the inter-frame motion information into the generated stego-image $\hat{I}_0$. The visualization results are shown in Figure 6, where we present the stego-image $\hat{I}_0$ containing motion information, the hidden features corresponding to $\hat{I}_0$, the overlay of I_0 and I_1 , and the residuals between $\hat{I}_0$ and I_0 . Most residual values fall within the range of 0 to 4, which corresponds to low bit residuals for uint8 images. Residual regions precisely highlight the locations of motion between frames, aligning with ghosting areas in the overlaid images and latent features. From this perspective, our method can be viewed as a low-bit steganographic technique, whereby motion information is embedded in the low bits of the downsampled results. Please note that we do not explicitly constrain how the motion information from high-frame-rate videos should be allocated and hidden into the low-frame-rate results. Instead, we encourage the model to optimize this process on its own. In more complex cases of non-integer frame rate conversion, the motion information may actually be dispersed across multiple downsampled frames. More experiments on frequency analysis and discussions on the robustness of steganography to compression are provided in the **supplementary file**.

5. Conclusions

In this paper, we tackle two critical challenges in video resampling. To preserve motion information during temporal downsampling, we develop the invertible motion steganography, which effectively embeds motion data from high-frame-rate videos into downsampled frames. To achieve flexible spatiotemporal resampling, we propose a 3D implicit modulation technique that allows for spatiotemporal resampling, including non-integer frame rate conversions. Our experiments demonstrate the effectiveness of these contributions across various video resampling tasks.

References

- [1] Kelvin C.K. Chan, Xintao Wang, Ke Yu, Chao Dong, and Chen Change Loy. BasicVSR: The search for essential components in video super-resolution and beyond. In *Conference on Computer Vision and Pattern Recognition*, pages 4947–4956, 2021. 4
- [2] Kelvin CK Chan, Shangchen Zhou, Xiangyu Xu, and Chen Change Loy. BasicVSR++: Improving video super-resolution with enhanced propagation and alignment. In *Conference on Computer Vision and Pattern Recognition*, pages 5972–5981, 2022. 4, 6
- [3] Yinbo Chen, Sifei Liu, and Xiaolong Wang. Learning continuous image representation with local implicit image function. In *International Conference on Computer Vision*, pages 8628–8638, 2021. 2, 4
- [4] Yi-Hsin Chen, Si-Cun Chen, Yen-Yu Lin, and Wen-Hsiao Peng. MoTIF: Learning motion trajectories with local implicit neural functions for continuous space-time video super-resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 23131–23141, 2023. 2, 6, 7
- [5] Zeyuan Chen, Yinbo Chen, Jingwen Liu, Xingqian Xu, Vidit Goel, Zhangyang Wang, Humphrey Shi, and Xiaolong Wang. VideoINR: Learning video implicit neural representation for continuous space-time super-resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 2047–2057, 2022. 2, 7
- [6] Ho Kei Cheng and Alexander G. Schwing. Xmem: Long-term video object segmentation with an atkinson-shiffrin memory model. In *European Conference on Computer Vision*, pages 640–658. 1
- [7] Ka Leong Cheng, Yueqi Xie, and Qifeng Chen. IICNet: A generic framework for reversible image conversion. In *International Conference on Computer Vision*, pages 1971–1980, 2021. 3
- [8] Xianhang Cheng and Zhenzhong Chen. Multiple video frame interpolation via enhanced deformable separable convolution. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7029–7045, 2022. 2, 6
- [9] Fayaz Ali Dharejo, Iyyakutti Iyappan Ganapathi, Muhammad Zawish, Basit Alawode, Moath Alathbah, Naoufel Werghi, and Sajid Javed. Swinwave-sr: Multi-scale lightweight underwater image super-resolution. *Information Fusion*, 103:102127, 2024. 4
- [10] Guanchen Ding and Chang Wen Chen. Towards omniscient feature alignment for video rescaling. In *International Conference on Acoustics, Speech and Signal Processing*, pages 4190–4194, 2024. 1
- [11] Laurent Dinh, David Krueger, and Yoshua Bengio. NICE: non-linear independent components estimation. In *International Conference on Learning Representations*, 2015. 5
- [12] Zhicheng Geng, Luming Liang, Tianyu Ding, and Ilya Zharkov. RSTT: real-time spatial temporal transformer for space-time video super-resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 17420–17430, 2022. 6
- [13] Mengxi Guo, Shijie Zhao, Yue Li, Junlin Li, Li Zhang, and Yue Wang. Invertible single image rescaling via steganography. In *International Conference on Multimedia and Expo*, pages 1–6, 2022. 3
- [14] Chengrui Huang, Wanjie Sun, and Zhenzhong Chen. Learned scale-arbitrary image downscaling for non-learnable upscaling. *IEEE Signal Processing Letters*, 30: 264–268, 2023. 1
- [15] Yan-Cheng Huang, Yi-Hsin Chen, Cheng-You Lu, Hui-Po Wang, Wen-Hsiao Peng, and Ching-Chun Huang. Video rescaling networks with joint optimization strategies for downscaling and upscaling. In *Conference on Computer Vision and Pattern Recognition*, pages 3527–3536. 1, 2
- [16] Zhewei Huang, Tianyuan Zhang, Wen Heng, Boxin Shi, and Shuchang Zhou. Real-time intermediate flow estimation for video frame interpolation. In *European Conference on Computer Vision*, pages 624–642, 2022. 2, 6
- [17] Shoko Imaizumi and Kei Ozawa. Multibit embedding algorithm for steganography of palette-based images. In *Image and Video Technology*, pages 99–110, 2013. 3
- [18] Meiguang Jin, Zhe Hu, and Paolo Favaro. Learning to extract flawless slow motion from blurry videos. In *Conference on Computer Vision and Pattern Recognition*, pages 8112–8121, 2019. 2, 5
- [19] Neil F. Johnson and Sushil Jajodia. Exploring steganography: Seeing the unseen. *Computer*, 31(2):26–34, 1998. 3, 8
- [20] Tarun Kalluri, Deepak Pathak, Manmohan Chandraker, and Du Tran. FLAVR: flow-agnostic video representations for fast frame interpolation. In *Winter Conference on Applications of Computer Vision*, pages 2070–2081, 2023. 6
- [21] Heewon Kim, Myungsub Choi, Bee Lim, and Kyoung Mu Lee. Task-aware image downscaling. In *European conference on computer vision*, pages 399–414, 2018. 2
- [22] Lingtong Kong, Boyuan Jiang, Donghao Luo, Wenqing Chu, Xiaoming Huang, Ying Tai, Chengjie Wang, and Jie Yang. IFRNet: Intermediate feature refine network for efficient frame interpolation. In *Conference on Computer Vision and Pattern Recognition*, pages 1969–1978, 2022. 2, 6
- [23] Wei-Sheng Lai, Jia-Bin Huang, Narendra Ahuja, and Ming-Hsuan Yang. Deep laplacian pyramid networks for fast and accurate super-resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 624–632, 2017. 5
- [24] Jaewon Lee and Kyong Hwan Jin. Local texture estimator for implicit representation function. In *International Conference on Computer Vision*, pages 1919–1928, 2022. 2, 4
- [25] Shang Li, Guixuan Zhang, Zhengxiong Luo, Jie Liu, Zhi Zeng, and Shuwu Zhang. Approaching the limit of image rescaling via flow guidance. In *British Machine Vision Conference*, pages 1–13. 2
- [26] Yue Li, Dong Liu, Houqiang Li, Li Li, Zhu Li, and Feng Wu. Learning a convolutional neural network for image compact-resolution. *IEEE Transactions on Image Processing*, 28(3): 1092–1107, 2019. 2
- [27] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. Swinir: Image restoration using swin transformer. In *International Conference on Computer Vision*, pages 1833–1844, 2021. 4

- [28] Ce Liu and Deqing Sun. A bayesian approach to adaptive video super resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 209–216, 2011. 5
- [29] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video swin transformer. In *Conference on Computer Vision and Pattern Recognition*, pages 3192–3201, 2022. 4
- [30] Yaqian Long, Xun Wang, Meng Xu, Shuyu Zhang, Shuguo Jiang, and Sen Jia. Dual self-attention swin transformer for hyperspectral image super-resolution. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–12, 2023. 4
- [31] Liying Lu, Ruizheng Wu, Huaijia Lin, Jiangbo Lu, and Jiaya Jia. Video frame interpolation with transformer. In *Conference on Computer Vision and Pattern Recognition*, pages 3532–3542, 2022. 6
- [32] Anish Mittal, Rajiv Soundararajan, and Alan C. Bovik. Making a “completely blind” image quality analyzer. *IEEE Signal Processing Letter*, 20(3):209–212, 2013. 8
- [33] Didrik Nielsen, Priyank Jaini, Emiel Hoogeboom, Ole Winther, and Max Welling. Survae flows: Surjections to bridge the gap between vaes and flows. In *Advances in Neural Information Processing Systems*, 2020. 3
- [34] Michiharu Niimi, Richard O. Eason, Hideki Noda, and Eiji Kawaguchi. High capacity and secure digital steganography to palette-based images. In *International Conference on Image Processing*, pages 917–920, 2002. 3
- [35] Zhihong Pan, Baopu Li, Dongliang He, Mingde Yao, Wenhao Wu, Tianwei Lin, Xin Li, and Errui Ding. Towards bidirectional arbitrary image rescaling: Joint optimization and cycle idempotence. In *Conference on Computer Vision and Pattern Recognition*, pages 17368–17377, 2022. 2
- [36] Zhihong Pan, Baopu Li, Dongliang He, Wenhao Wu, and Errui Ding. Effective invertible arbitrary image rescaling. In *Winter Conference on Applications of Computer Vision*, pages 5405–5414, 2023. 2
- [37] Kuldeep Purohit, Anshul B. Shah, and A. N. Rajagopalan. Bringing alive blurred moments. In *Conference on Computer Vision and Pattern Recognition*, pages 6830–6839, 2019. 2, 5
- [38] Chenyang Qi, Xin Yang, Ka Leong Cheng, Ying-Cong Chen, and Qifeng Chen. Real-time 6k image rescaling with rate-distortion optimization. In *Conference on Computer Vision and Pattern Recognition*, pages 14092–14101, 2023. 1, 2
- [39] E. Shechtman, Y. Caspi, and M. Irani. Space-time super-resolution. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(4):531–545, 2005. 2
- [40] Haichao Shi, Jing Dong, Wei Wang, Yinlong Qian, and Xiaoyu Zhang. SSGAN: secure steganography based on generative adversarial networks. In *Advances in Multimedia Information Processing*, pages 534–544, 2017. 3
- [41] Hyeonjun Sim, Jiyoung Oh, and Munchurl Kim. XVFI: extreme video frame interpolation. In *International Conference on Computer Vision*, pages 14469–14478, 2021. 6
- [42] Shuochen Su, Mauricio Delbracio, Jue Wang, Guillermo Sapiro, Wolfgang Heidrich, and Oliver Wang. Deep video deblurring for hand-held cameras. In *Conference on Computer Vision and Pattern Recognition*, pages 1279–1288, 2017. 5
- [43] Wanjie Sun and Zhenzhong Chen. Learned image downscaling for upscaling using content adaptive resampler. *IEEE Transactions on Image Processing*, 29:4027–4040, 2020. 1
- [44] Xin Tao, Hongyun Gao, Renjie Liao, Jue Wang, and Jiaya Jia. Detail-revealing deep video super-resolution. In *International Conference on Computer Vision*, pages 4472–4480, 2017. 5, 7
- [45] Yuan Tian, Guo Lu, Xiongkuo Min, Zhaohui Che, Guangtao Zhai, Guodong Guo, and Zhiyong Gao. Self-conditioned probabilistic learning of video rescaling. In *International Conference on Computer Vision*, pages 4470–4479, 2021. 1, 2
- [46] Hai Wang, Xiaoyu Xiang, Yapeng Tian, Wenming Yang, and Qingmin Liao. STDAN: deformable attention network for space-time video super-resolution. *IEEE Transactions on Neural Networks and Learning Systems Information for Authors*, pages 10606–10616, 2024. 6
- [47] Dafeng Wei, Ye Tian, Liqing Wei, Hong Zhong, Siqian Chen, Shiliang Pu, and Hongtao Lu. Efficient dual attention slowfast networks for video action recognition. *Computer Vision and Image Understanding*, 222:103484, 2022. 1
- [48] Da-Chun Wu and Wen-Hsiang Tsai. A steganographic method for images by pixel-value differencing. *Pattern Recognition Letters*, 24(9-10):1613–1626, 2003. 3
- [49] H-C Wu, N-I Wu, C-S Tsai, and M-S Hwang. Image steganographic scheme based on pixel-value differencing and lsb replacement methods. *IEE Proceedings-Vision, Image and Signal Processing*, 152(5):611–615, 2005. 3
- [50] Xiaoyu Xiang, Yapeng Tian, Vijay Rengarajan, Lucas D. Young, Bo Zhu, and Rakesh Ranjan. Learning spatio-temporal downsampling for effective video upscaling. In *European Conference on Computer Vision*, pages 162–181. 2, 5, 6
- [51] Xiaoyu Xiang, Yapeng Tian, Yulun Zhang, Yun Fu, Jan P Allebach, and Chenliang Xu. Zooming slow-mo: Fast and accurate one-stage space-time video super-resolution. In *Conference on Computer Vision and Pattern Recognition*, pages 3370–3379, 2020. 6
- [52] Mingqing Xiao, Shuxin Zheng, Chang Liu, Yaolong Wang, Di He, Guolin Ke, Jiang Bian, Zhouchen Lin, and Tie-Yan Liu. Invertible image rescaling. In *European Conference on Computer Vision*, pages 126–144. 1, 2
- [53] Mingqing Xiao, Shuxin Zheng, Chang Liu, Zhouchen Lin, and Tie-Yan Liu. Invertible rescaling network and its extensions. *International Journal of Computer Vision*, 131(1): 134–159, 2023. 2
- [54] Jinbo Xing, Wenbo Hu, Menghan Xia, and Tien-Tsin Wong. Scale-arbitrary invertible image downscaling. *IEEE Transactions on Image Processing*, 32:4259–4274, 2023. 1, 2
- [55] Youmin Xu, Chong Mou, Yujie Hu, Jingfen Xie, and Jian Zhang. Robust invertible image steganography. In *Conference on Computer Vision and Pattern Recognition*, pages 7865–7874, 2022. 3
- [56] T. Xue, B. Chen, J. Wu, D. Wei, and W. T. Freeman. Video enhancement with task-oriented flow. *International Journal of Computer Vision*, 127(8):1106–1125, 2019. 5
- [57] Yu-Qi Yang, Yu-Xiao Guo, Jian-Yu Xiong, Yang Liu, Hao Pan, Peng-Shuai Wang, Xin Tong, and Baining Guo.

- Swin3d: A pretrained transformer backbone for 3d indoor scene understanding. *arXiv preprint arXiv:2304.06906*, 2023. 4
- [58] Guozhen Zhang, Yuhan Zhu, Haonan Wang, Youxin Chen, Gangshan Wu, and Limin Wang. Extracting motion and appearance via inter-frame attention for efficient video frame interpolation. In *Conference on Computer Vision and Pattern Recognition*, pages 5682–5692, 2023. 2, 6
 - [59] Junxi Zhang, Ding Ding, Zhenzhong Chen, and Shan Liu. [FCM] CE 2.3.1 Temporal Resampling. In *ISO/IEC JTC 1/SC29/WG4 M67615*, Rennes, 2024. 1
 - [60] Kaihao Zhang, Wenhan Luo, Björn Stenger, Wenqi Ren, Lin Ma, and Hongdong Li. Every moment matters: Detail-aware networks to bring a blurry image alive. In *ACM International Conference on Multimedia*, pages 384–392. 2, 5
 - [61] Kevin Alex Zhang, Alfredo Cuesta-Infante, Lei Xu, and Kalyan Veeramachaneni. Steganogan: High capacity image steganography with gans. *arXiv preprint arXiv:1901.03892*, 2019. 3
 - [62] Yulun Zhang, Kunpeng Li, Kai Li, Lichen Wang, Bineng Zhong, and Yun Fu. Image super-resolution using very deep residual channel attention networks. In *European Conference on Computer Vision*, pages 294–310, 2018. 4
 - [63] Yuantong Zhang, Huairui Wang, Han Zhu, and Zhenzhong Chen. Optical flow reusing for high-efficiency space-time video super resolution. *IEEE Transactions on circuits and systems for video technology*, 33(5):2116–2128, 2023. 6
 - [64] Yuantong Zhang, Daiqin Yang, Zhenzhong Chen, and Wenpeng Ding. Continuous space-time video super-resolution with multi-stage motion information reorganization. *ACM Transactions on Multimedia Computing, Communications and Applications*, early access, 2024. 2, 6
 - [65] Yuantong Zhang, Hanyou Zheng, Daiqin Yang, Zhenzhong Chen, Haichuan Ma, and Wenpeng Ding. Space-time video super-resolution with neural operator. *arXiv preprint arXiv:2404.06036*, 2024. 2, 7